

Homework 8

40 points

1. For the given function, perform 3 steps of gradient descent. Estimate what value does x approach as the iterations continue. **2+2+2+1= 7 points**

$$f(x) = (x - 4)^2$$

$$f'(x) = 2(x - 4)$$

$$x_{k+1} = x_k - \alpha f'(x_k) = x_k - 2\alpha(x_k - 4) = (1 - 2\alpha)x_k + 8\alpha$$

$$x_1 = (1 - 2\alpha)x_0 + 8\alpha = 4 + (1 - 2\alpha)(x_0 - 4)$$

$$x_2 = 4 + (1 - 2\alpha)^2(x_0 - 4)$$

$$x_3 = 4 + (1 - 2\alpha)^3(x_0 - 4)$$

Behavior:

If $0 < \alpha < 1$, then $|1 - 2\alpha| < 1$ and $x_k \rightarrow 4$ (the minimum)

If $\alpha \geq 1$ it diverges, if $1/2 < \alpha < 1$ also oscillates around 4 while diverging.

2. Given $f(x) = (x - 2)^2$. Compare the results of one step of gradient descent from $x_0 = 0$ for $\alpha = 0.1$ and $\alpha = 1.5$. Which case "overshoots" the minimum? **2+2+1=5 points**

$$f'(x) = 2(x-2)$$

$$\text{One step: } x_1 = x_0 - \alpha 2(x_0 - 2) = 0 - 2\alpha(-2) = 4\alpha$$

$$\text{For } \alpha = 0.1 \quad x_1 = 0.4$$

$$\text{For } \alpha = 1.5 \quad x_1 = 6$$

The case with $\alpha = 1.5$ overshoots the minimum

3. Evaluate the following integrals: **5*5 = 25 points**

- $\int (2x + 3)^5 dx$
- $\int \frac{1}{\sqrt{1-x^2}} dx$
- $\int e^x \cos x dx$
- $\int_1^4 \left(\frac{8}{\sqrt{t}} - 12\sqrt{t^3} \right) dt$
- $\int_{-4}^{-1} \left(\sqrt[3]{5-2y} + \frac{7}{(5-2y)} \right) dy$

a. $\int (2x+3)^5 dx = \frac{(2x+3)^6}{12} + C$

b. $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$

c. $\int e^x \cos x dx = \frac{e^x}{2} (\sin x + \cos x) + C$

d. $\int_1^4 \left(\frac{8}{\sqrt{t}} - 12\sqrt{t^3} \right) dt$
 $= \int_1^4 (8t^{-1/2} - 12t^{3/2})$

Antiderivative: $16t^{1/2} - \frac{24}{5} t^{5/2}$

$$(16 \times 2 - \frac{24}{5} \times 32) - (16 \times 1 - \frac{24}{5} \times 1) = \boxed{-132 - 8}$$

e. $\int_{-4}^{-1} \left(\sqrt[3]{5-2y} + \frac{7}{5-2y} \right) dy$
 $u = 5-2y \quad du = -2dy \quad dy = -\frac{du}{2}$

$$\int_{-4}^{-1} \left(u^{1/3} + \frac{7}{u} \right) \left(-\frac{du}{2} \right) = -\frac{1}{2} \int_{13}^7 \left(u^{1/3} + \frac{7}{u} \right) du = \frac{1}{2} \int_7^{13} \left(u^{1/3} + \frac{7}{u} \right) du$$

$$\int u^{1/3} du = \frac{3}{4} u^{4/3} \quad \int \frac{7}{u} du = 7 \ln|u|$$

$$\left[\frac{3}{4} (13)^{4/3} + 7 \ln 13 \right] - \left[\frac{3}{4} (7)^{4/3} + 7 \ln 7 \right] = \frac{3}{4} (13^{4/3} - 7^{4/3}) + 7(\ln 13 - \ln 7)$$

$$\frac{1}{2} \left[\frac{3}{4} (13^{4/3} - 7^{4/3}) + 7 \ln \frac{13}{7} \right] = \frac{3}{8} (13^{4/3} - 7^{4/3}) + \frac{7}{2} \ln \frac{13}{7} = \boxed{8.60795}$$

4. The velocity of a particle is $v(t) = 3t^2 - 2t + 1$ m/sec. Find the displacement from $t = 0$ to $t = 4$. **3 points**

$$\int_0^4 (3t^2 - 2t + 1) dt = [t^3 - t^2 + t]_0^4 = 64 - 16 + 4 = \boxed{52}$$